

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS1

SUBJECT: Agricultural Science

TOPIC: Importance & Branches of Agriculture

Introduction

- Agriculture is the cultivation of crops and the rearing of animals for man's use. It is the oldest occupation of man and remains the backbone of the Nigerian economy.

Importance of agriculture

- Provides food (carbohydrates, proteins, vitamins) for the population.
- Supplies raw materials to industries (cotton for textiles, oil palm for soap, cocoa for chocolate, sugarcane for sugar).
- Provides employment for a large proportion of the labour force, especially in rural areas.
- Earns foreign exchange through export crops (cocoa, rubber, palm oil, groundnut) and saves foreign exchange through import substitution.
- Provides a market for industrial products and capital for development.

Branches of agriculture

- **Agronomy/crop production** — cultivation of arable crops.
- **Animal husbandry** — rearing of farm animals.
- **Agricultural economics** — the business side of farming.
- **Agricultural extension** — educating farmers.
- **Agricultural engineering** — farm machines and structures.
- **Fisheries** — fish farming; **forestry** — trees and timber; **horticulture** — vegetables, fruits and flowers.

CLASS EVALUATION

1. Define agriculture.
2. Mention three importance of agriculture to the Nigerian economy.
3. Name four branches of agriculture.
4. How does agriculture earn foreign exchange for Nigeria?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

CLASS: SS1

SUBJECT: Agricultural Science

TOPIC: Agricultural Ecology

Introduction

- Ecology is the study of the relationships between living things and their environment. Agricultural ecology examines how farm crops and animals interact with their surroundings.

Components of the environment

- **Abiotic factors** — non-living: sunlight, temperature, rainfall, wind, soil, water, air and topography.
- **Biotic factors** — living: plants, animals, micro-organisms (bacteria, fungi), and man, who influences the farm through cultivation, fertilisers, irrigation and pest control.

Terms in ecology

- **Habitat** — the natural home of an organism; **population** — all members of one species in a habitat; **community** — all the populations living together; **ecosystem** — the community plus its non-living environment, e.g. a pond, a forest, a farmland; **niche** — the role of an organism in its habitat.

Adaptation on the farm

- Crops adapt to day length (photoperiodism), temperature and moisture; animals adapt to climate (trypanotolerant breeds in tsetse-infested areas). Understanding ecology helps the farmer choose the right crop for the right place — land suitability.

CLASS EVALUATION

1. Distinguish between abiotic and biotic factors.
2. Define ecosystem with an example.
3. What is a niche?
4. State two ways ecology helps a farmer.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS1

SUBJECT: Agricultural Science

TOPIC: Soil Formation

Introduction

- Soil is the loose, weathered surface material of the earth in which plants grow. It is formed from the breakdown of parent rocks over long periods — this process is soil formation (pedogenesis).

Processes of soil formation

- **Weathering** — physical (temperature, frost, wind), chemical (oxidation, hydration, carbonation) and biological (roots, earthworms) breakdown of the parent rock.
- **Decomposition** — humus is formed as dead plants and animals decay.
- **Leaching** — rainwater carries soluble minerals down the profile; **translocation** — movement of clay and humus within the soil.

Factors of soil formation

- Parent material — determines the minerals; climate — heat and rainfall speed up weathering; organisms — plants, animals and microbes add humus; topography — slope affects drainage and erosion; time — soils need hundreds of years to mature.

The soil profile

- From the top down: **A-horizon** (topsoil, dark with humus, most fertile), **B-horizon** (subsoil, mineral-rich, clay accumulation), **C-horizon** (weathered parent material) and **bedrock** (unweathered rock).

CLASS EVALUATION

1. Define soil.
2. List the factors of soil formation.
3. Explain leaching.
4. Name the horizons of the soil profile from top to bottom.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS1

SUBJECT: Agricultural Science

TOPIC: Soil Properties

Introduction

- The properties of a soil determine what it can grow. They include physical properties (texture, structure, porosity, colour) and chemical properties (pH, organic matter, nutrient content).

Texture and structure

- **Texture** — the proportions of sand, silt and clay particles. Sandy soils are loose, well-drained, warm and easy to cultivate but poor in nutrients; clay soils are sticky, hold water and nutrients but are poorly drained and hard to work; loam is the ideal mixture for most crops.
- **Structure** — the arrangement of soil particles into crumbs or blocks; good crumb structure allows water and air movement.

Porosity, permeability and capillarity

- Porosity is the space between soil particles (pores); permeability is the ease with which water passes through; capillarity is the upward movement of water in the soil. Sandy soils have large pores (high permeability, low capillarity); clay soils have small pores (low permeability, high capillarity).

Soil pH and fertility

- pH 7 is neutral; below 7 is acidic, above 7 alkaline. Most crops prefer pH 6–7. Lime raises pH; organic matter and fertilisers add nutrients (nitrogen, phosphorus, potassium). A fertile soil is rich in humus, balanced in nutrients and well aerated.

CLASS EVALUATION

1. Define soil texture.
2. Which soil type retains water best and why?
3. What is soil pH, and how is an acidic soil corrected?
4. Describe loam.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS1

SUBJECT: Agricultural Science

TOPIC: Simple Farm Tools & Maintenance

Introduction

- Farm tools are implements used in farm operations. Simple hand tools are used on small farms; their proper use and maintenance prevent accidents and lengthen their life.

Common tools and their uses

- **Cutlass** — clearing land, cutting stems; **hoe** — weeding, loosening soil, making ridges; **spade** — digging and turning soil; **shovel** — lifting sand/gravel; **rake** — gathering trash, levelling; **mattock** — heavy digging, prying; **garden fork** — loosening compacted soil; **watering can** — watering seedlings; **hand trowel** — transplanting; **sickle** — harvesting grains; **sprinkler and knapsack sprayer** — applying water and chemicals.

Maintenance of tools

- Clean tools after every use and oil metal parts against rust.
- Sharpen blades (cutlass, hoe) regularly and store in a dry place.
- Replace worn or broken handles; never leave tools in the rain or on wet soil.
- Use each tool for its intended purpose only, and keep tools under lock to prevent theft.

CLASS EVALUATION

1. Name five simple farm tools and state one use of each.
2. How do you care for a cutlass?
3. Why is a hoe not used for lifting gravel?
4. State two safety practices when using farm tools.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: SIX

CLASS: SS1

SUBJECT: Agricultural Science

TOPIC: Agricultural Systems & Farm Records

Introduction

- Farm systems are the methods farmers use to organise cropping and animal rearing: subsistence, commercial, mixed, pastoral, plantation and mechanised farming.

Systems of farming

- **Subsistence** — food for the family, small scale, simple tools.
- **Commercial** — large scale, for sale, uses improved inputs.
- **Mixed farming** — crops and animals together, waste from one feeds the other.
- **Mixed cropping** — two or more crops on the same piece of land at the same time.
- **Pastoral** — animals only (nomadic, ranching); **plantation** — a single cash crop on a large estate (oil palm, rubber, cocoa); **shifting cultivation/bush fallowing** — land is cropped and rested; **mechanised** — machines replace human labour.

Farm records

- Farmers keep records for planning and control: **inventory** (assets and stock), **production record** (output), **labour record**, **sales and purchase record**, **financial/cash book** (income and expenditure), and **profit and loss**. Records show whether the farm is making a profit and guide future decisions.

CLASS EVALUATION

1. Differentiate between subsistence and commercial farming.
2. What is mixed farming and why is it advantageous?
3. List four farm records a farmer should keep.
4. Why is shifting cultivation no longer fashionable?